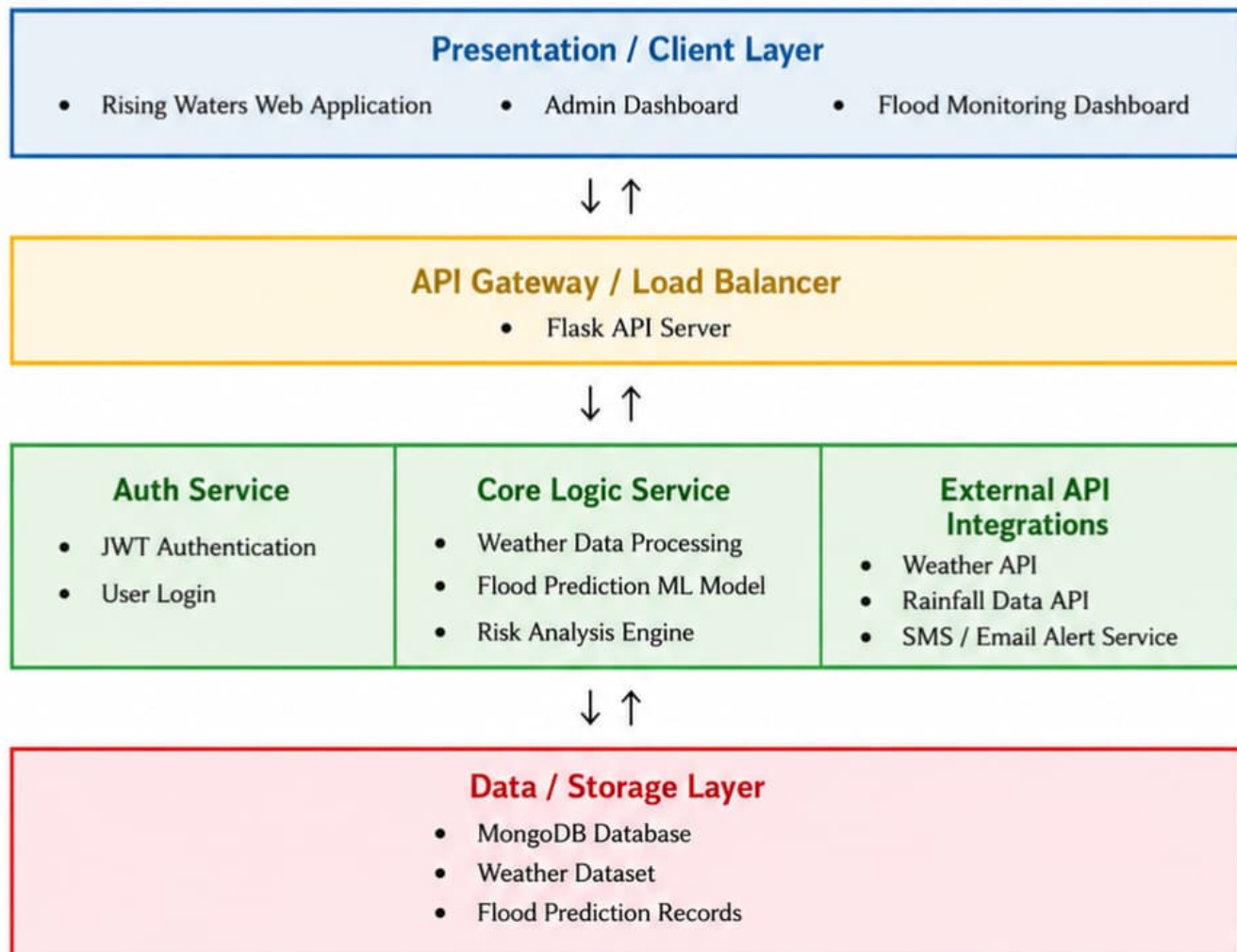


Solution Architecture

Date	2 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for disaster management officers to input weather data, view flood predictions, risk maps, and alerts through the Rising Waters web application and dashboards.	HTML, CSS, JavaScript, Bootstrap, Chart.js
API Gateway	Acts as a single entry point for all client requests and routes them securely to the appropriate services.	Flask (REST API), Nginx (Reverse Proxy)
Microservices	Handles core business logic including weather data processing, flood prediction using machine learning models, risk analysis, and report generation.	Python, Scikit-learn, Pandas, NumPy, Joblib
Database	Stores weather data, user information, prediction results, alert logs, and other system records for retrieval and analysis.	MongoDB